POSTS AND TELECOMMUNICATIONS
INSTITUE OF TECHNOLOGY
INFORMATION TECHNOLOGY

INTERNSHIP

REPORT

WoxionChat

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Table of contents

1 Technical Accomplishments **2**
1.1 AI-Centric Chat Interface 2
1.2 Administrative Dashboard (The Boss UI) 2
1.3 Responsive Design & State Management 2

2 Technology Stack Update **2**

3 Progress Evaluation **3**

Table of contents

1 Technical Accomplishments

1.1 AI-Centric Chat Interface

The core communication interface was designed and implemented with a focus on user experience:

- **Real-time Streaming UI:** Integrated WebSocket listeners to render AI responses token-by-token, ensuring a fluid and interactive experience.
- **Markdown & Syntax Highlighting:** Implemented full support for rich-text responses, including code blocks, tables, and mathematical formulas (via LaTeX-style rendering).
- **Agentic Thought-Trace:** Developed a "Thought Process" toggle component that allows users to view the AI Agent's reasoning steps and tool-calling actions in a transparent, expandable view.

1.2 Administrative Dashboard (The Boss UI)

A specialized interface was built for administrative oversight:

- **Session Monitoring:** Created a high-level overview map and list view for "The Boss" to monitor active `ip_address` locations and `user_agent` profiles in real-time.
- **User Management Portal:** Developed the UI for managing `is_staff` and `is_superuser` flags, including a granular permission editor for employee roles.
- **Audit Logs:** Built a searchable log viewer to track critical actions performed by both users and AI Agents.

1.3 Responsive Design & State Management

- **Fluid Layouts:** Utilized Tailwind CSS to ensure the platform is fully responsive across mobile, tablet, and desktop orientations.
- **Persistent Context:** Implemented local state synchronization with the backend to ensure that long-form conversations remain consistent across page refreshes.

2 Technology Stack Update

- **Frontend Framework:** React with Tailwind CSS for rapid UI development and styling.
- **State Management:** Redux Toolkit / React Context for handling complex session and AI states.
- **Icons & Assets:** Lucide-React for consistent iconography across the dashboard.
- **API Integration:** Axios for RESTful calls and native WebSocket API for real-time streaming.

3 Progress Evaluation

- **Status:** On Track.
- **Observations:** The implementation of the "Thought-Trace" UI has significantly increased user trust during complex agentic tasks. The frontend is successfully consuming the Django AI API with minimal latency.